Near-infrared optical properties of *ex vivo* human skin and subcutaneous tissues measured using the Monte Carlo inversion technique

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Abstract. The absorption and transport scattering coefficients of caucasian and negroid dermis, subdermal fat and muscle have been measured for all wavelengths between 620 and 1000 nm. Samples of tissue 2 mm thick were measured *ex vivo* to determine their reflectance and transmittance. A Monte Carlo model of the measurement system and light transport in tissue was then used to recover the optical coefficients. The sample reflectance and transmittance were measured using a single integrating sphere 'comparison' method. This has the advantage over conventional double-sphere techniques in that no corrections are required for sphere properties, and so measurements sufficiently accurate to recover the absorption coefficient reliably could be made. The optical properties of caucasian dermis were found to be approximately twice those of the underlying fat layer. At 633 nm, the mean optical properties over 12 samples were 0.033 mm^{-1} and 0.013 mm^{-1} for absorption coefficient and 2.73 mm^{-1} and 1.26 mm^{-1} for transport scattering coefficient for caucasian dermis and the underlying fat layer respectively. The transport scattering coefficient for all biological samples showed a monotonic decrease with increasing wavelength. The method was calibrated using solid tissue phantoms and by comparison with a temporally resolved technique.

1. Introduction

The optical properties of the skin are of interest because of their effect on non-invasive optical measurements of deeper tissue, and also because of the possibility of using the skin as an accessible organ for determining the constituents of blood *in vivo*. The difficulties involved in determining the optical properties of tissue *in vivo* are well known: advanced instrumentation is required (time or phase resolved spectroscopy) and the volume of tissue probed is hard to define. However, it has been shown (Delpy *et al* 1988) that the differential pathlength of light in tissue measured *post-mortem* is similar to that measured *in vivo*. Tissue excision and storage may produce changes in the optical properties due to blood drainage but these effects are predictable as a loss of absorption coefficient, and *ex vivo* measurements remain a good method of isolating particular tissue types.

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The purpose of the method described here is to obtain the average optical properties of small volumes of tissue (less than 0.5 ml). These optical properties are intended to be used in mathematical models of light transport in larger heterogeneous tissue volumes such as the head, limbs or abdomen for optical imaging and spectroscopy applications. Mathematical models of large complex tissue volumes tend to use finite element methods and approximations to the radiation transport equation (for example Arridge *et al* 1993). An accuracy of 10% in optical properties is typically sufficient to evaluate the effects of light transport across tissue boundaries. This degree of accuracy is also consistent with ‘normal’ variations of absorption coefficient caused by blood volume variations and what appears to be the natural variance of scattering coefficient seen in tissues such as the brain (Matcher *et al* 1997).

Tissue optical properties are usually described in terms of three parameters: the scattering coefficient μ_s, absorption coefficient μ_a and scattering anisotropy factor g. Optical measurements on tissue, such as reflectance, are the result of the combined effect of these properties making it necessary to understand light transport. The light transport model compares its measurement predictions with experimental temporal, frequency and/or spatial measurements in order to recover these optical properties. Of these, steady-state intensity measurements are the simplest to perform. The radiative transport equation (Ishimaru 1989) is perhaps the most successful for predicting the flux for a particular geometry and set of optical coefficients. However, complications can occur at boundaries and not all geometries are trivial to compute analytically. Alternatively, stochastic models such as the Monte Carlo (Wilson and Adam 1983) and random walk models (Bonner *et al* 1987) may be used. A third alternative is the adding doubling method first applied to tissue optics by Prahl (1988). Previous investigators have used the Monte Carlo inversion technique (van der Zee 1992, Graaff *et al* 1993) and the inverse adding doubling method (Prahl *et al* 1993, Pickering *et al* 1993, Troy *et al* 1996).

The method employed here is to compare the experimentally measured diffuse reflectance and transmittance of dissected tissue samples with the results of a series of Monte Carlo models generated using known optical coefficients in the same sample geometry. The reflectance and transmittance of the tissue samples are measured using a single integrating sphere using a sample/reference comparison method (Jacquez and Kuppenheim 1950) which reduces the sources of error in measuring the diffuse reflectance and transmittance of the sample. The use of a spectrometer allows simultaneous detection over a 400 nm wavelength range. The measurements discussed in this paper were all carried out at room temperature (approximately 25 °C); the effect of temperature on tissue optical properties over the range 25–40 °C is examined in a companion paper (Laufer *et al* 1998).

2. Measurement method

2.1. Sample preparation

Human abdominal and breast tissue was obtained from plastic surgery or *post-mortem* examinations. All tissue was measured within five days of excision or death, and its condition was considered normal. Breast tissue was taken from the underside of the breast during mastectomy or breast reduction, and abdominal tissue from the mid-line incision in *post-mortem* examinations and a similar site during stomach reductions. Samples were refrigerated during storage, and allowed to return to room temperature before being dissected. The skin side of the samples were wiped with antiseptic tissues (‘WetOnes’, Jeyes, Norfolk, UK) to remove blood and surgical markings, and if necessary shaved. They

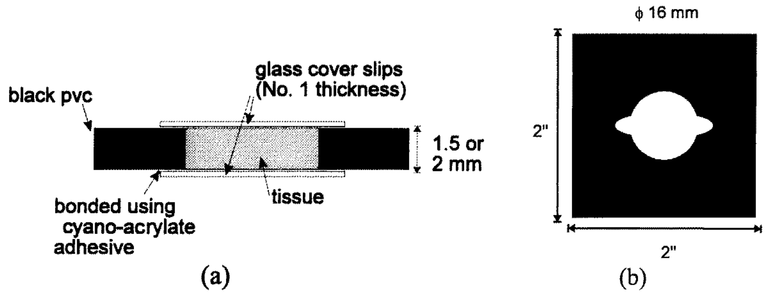

Figure 1. Design of tissue sample holder: (*a*) side and (*b*) plan view.

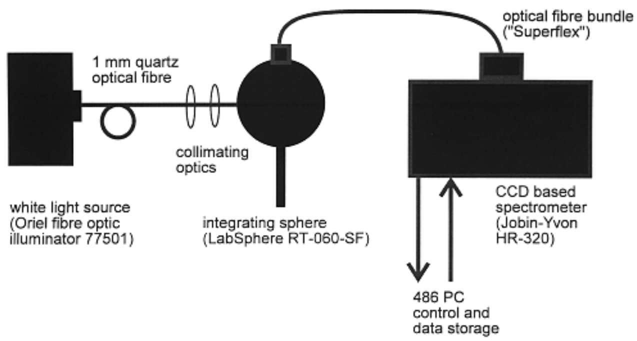

Figure 2. Schematic diagram of the integrating sphere apparatus.

were then separated into 'dermis' and 'subdermis' layers using a razor blade; dermis being all tissue from the skin surface to the bottom of the dermis (including epidermis, 1.5 to 2 mm thick) and subdermis the 2 mm of fat immediately below the dermis. Discs of tissue 16 mm in diameter were cut from the separated layers using a custom-made stainless steel punch and mounted in black PVC plastic holders between two glass coverslips, as shown in figure 1. The coverslips were fixed to the holders with cyano-acrylate adhesive so that the samples were completely enclosed for ease and safety of handling. This also prevented desiccation of the sample during the measurements. One sample of muscle was also recovered during the sample preparation, and mounted so that illumination was perpendicular to the direction of muscle fibres. For each subject, three subsamples of dermis and subdermis were mounted and measured.

2.2. Reflectance and transmittance measurements

The samples were mounted at an open port of a single integrating sphere containing no baffles (LabSphere RT-060-SF) and illuminated using an optical fibre-coupled white light source as shown in figure 2. The beam was collimated and its diameter was 5 ± 0.5 mm for both reflectance and transmittance measurements. The diffuse reflected or transmitted light was collected via a fibre bundle placed at the 'north pole' of the sphere and measured using a CCD-based spectrophotometer. Light over all wavelengths from 620 to 1000 nm could therefore be measured simultaneously. The sphere design allowed the use of the 'comparison' method for determining the reflectance or transmission of a sample. This technique has advantages (Jacquez and Kuppenheim, 1950) over the 'substitution' method

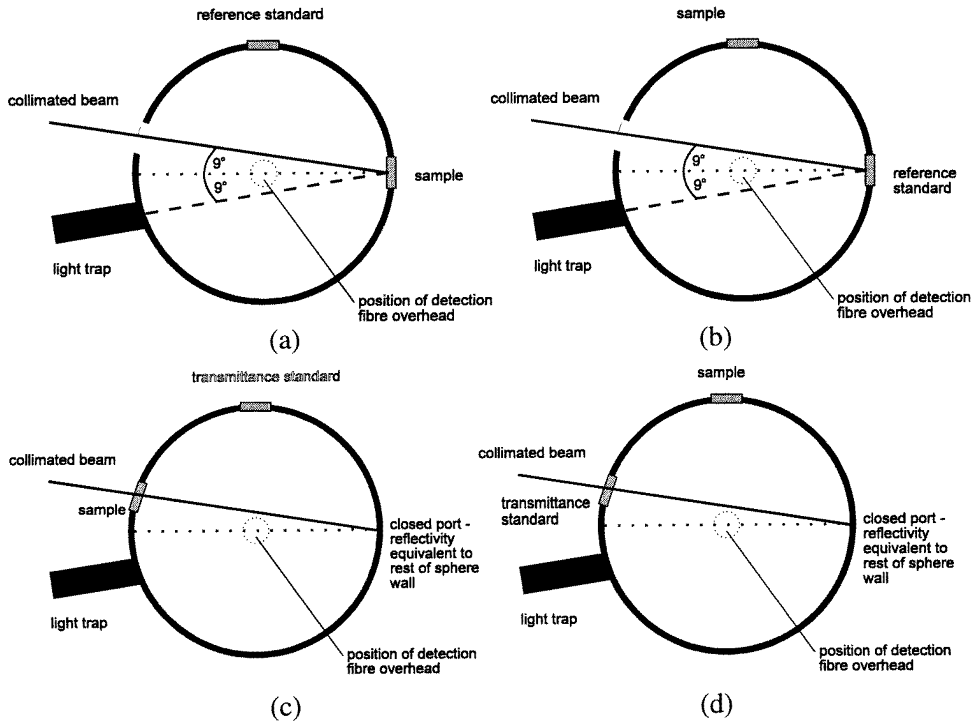

Figure 3. Integrating sphere configuration for (*a*) sample reflectance measurement, (*b*) reference standard reflectance measurement, (*c*) sample transmittance measurement and (*d*) reference standard transmittance measurement.

employed by van der Zee (1992) and Pickering *et al* (1993), as it requires no knowledge of the sphere geometry or properties.

A Spectralon 50% reflectance standard (LabSphere SRS-50-010) was used as a reference for the reflectance measurement, and a single coverslip was attached to the front with a thin layer of water to match the effects of the tissue mounting. The standard used for the transmittance measurements was an empty sample holder fronted with two coverslips, again attached with a slick of water, providing 100% transmittance after specular reflectance had been removed. A light trap was present in the integrating sphere to remove specular reflectance during the reflectance measurements. Figures 3(*a*) to (*d*) show the sphere configuration for the sample and reference standard reflectance and transmittance measurements.

The sample and reference standard remained on the sphere surface during all the measurements so that the sphere efficiency remained the same whether the sample or standard was illuminated. By conservation of energy therefore

$$\frac{I_{\text{sample}}}{I_{\text{ref}}} = \frac{R_{\text{sample}}}{R_{\text{ref}}} \tag{1}$$

where I_{sample} and I_{ref} are the detected intensities for sample and reference illumination respectively, R_{sample} is the reflectance of the sample and R_{ref} is the known reflectance of the standard. Similarly for transmittance measurements using a 100% reference standard

$$\frac{I_{\text{sample}}}{I_{\text{ref}}} = T_{\text{sample}}. \tag{2}$$

The transmittance and reflectance of each sample was measured twice, illuminating both sides of the sample in turn. This was particularly necessary for the dermis samples as the heterogeneous layered structure of the tissue produces very different reflectivities on opposite sides of the sample for wavelengths shorter than 800 nm.

3. Monte Carlo model

A Monte Carlo model of light interaction with tissue was used to simulate the sample geometry. The model takes four parameters: absorption coefficient, μ_a, scattering coefficient, μ_s, refractive index n and scattering phase function. For the purposes of this study, a Henyey–Greenstein scattering phase function was used, with the average cosine, g, set to 0.9. This is within the range of reported values for skin (Jacques *et al* 1987, Treweek and Barbenel 1996). The refractive index of the tissue was assumed to be $n = 1.4$ (Bolin *et al* 1989), although because the measurement method removes specularly reflected light, the model used $n = 1.0$ for light entering the tissue simulation. The value of $n = 1.4$ was only used when calculating the direction and proportion of light exiting the modelled tissue.

The tissue modelled had the same dimensions as the measured samples, 2 mm thick and 16 mm in diameter, with a 5 mm diameter Gaussian beam incident at 8° to the normal. Photons reaching the outer circumference of the sample were modelled as at an air/tissue boundary. The proportion of photons leaving the sample was dependent upon their angle of incidence and any photons exiting the sample were not readmitted. This is to mimic the experimental situation where light emitted from the sides of the tissue sample is absorbed by the black sample holders. (In fact, this assumption is an approximation—parts of the tissue samples were in contact with the sides of the sample holder and there will also be refractive effects for photons travelling from air into the sample holder. However, these two effects are in opposition. Repeating the Monte Carlo simulation for a perfect absorbing boundary produced results that were identical to those for the tissue/air interface to within the noise seen in figures 4 and 5.)

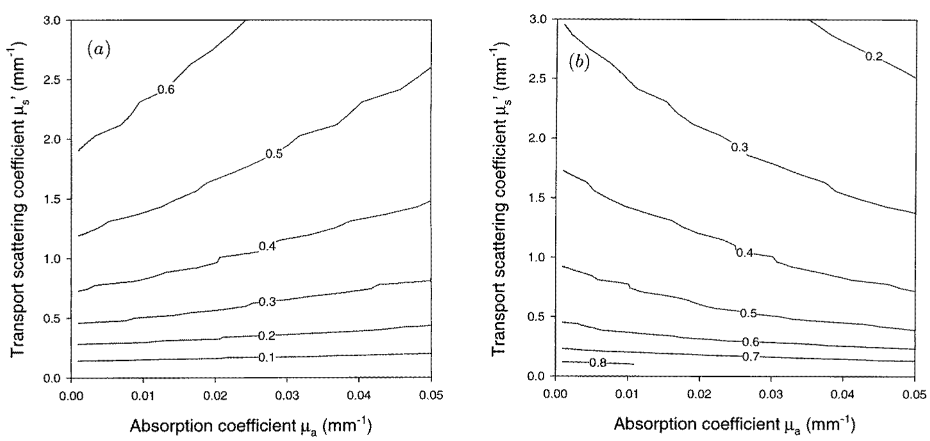

Figure 4. Results of Monte Carlo model: (*a*) reflectance and (*b*) transmittance as a function of μ_a and μ_s'.

Four thousand combinations of the absorption and scattering coefficients (μ_s = 1–75 mm^{-1}, $\mu_a = 0.001$–0.1 mm^{-1} for a fixed g of 0.9) were implemented using the model, and the resulting reflectance and transmittance values stored in a look-up table. The results of the sample measurements were then compared with the modelled data, and the absorption and transport scattering ($\mu_s' = \mu_s(1-g)$) coefficients of the tissue inferred. Values were linearly interpolated between the modelled coefficients to improve accuracy. The results of the simulations are plotted in figures 4(*a*) and (*b*), reflectance and transmittance as a function of the modelled optical properties, μ_a and μ_s'. Figures 5(*a*) and (*b*) show the inferred optical properties as a function of reflectance and transmittance. The Monte Carlo model is run for scattering and absorption values logarithmically distributed between the minimum and maximum values.

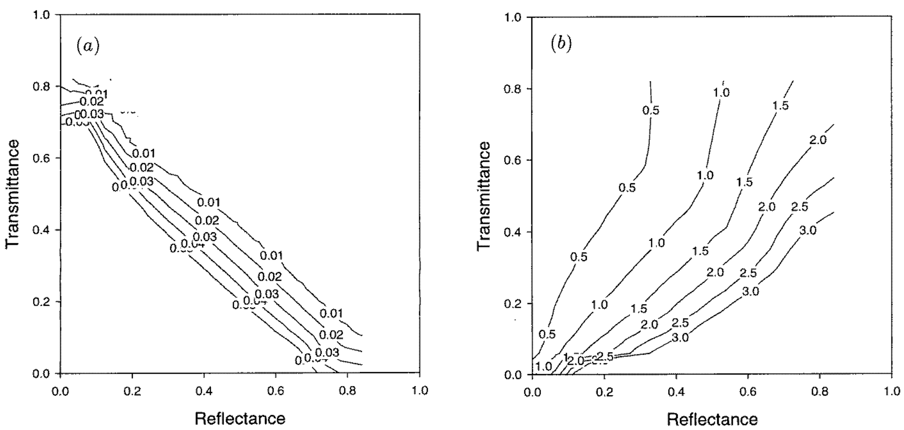

Figure 5. Inversion of Monte Carlo model: (*a*) μ_a and (*b*) μ_s' as a function of reflectance and transmittance.

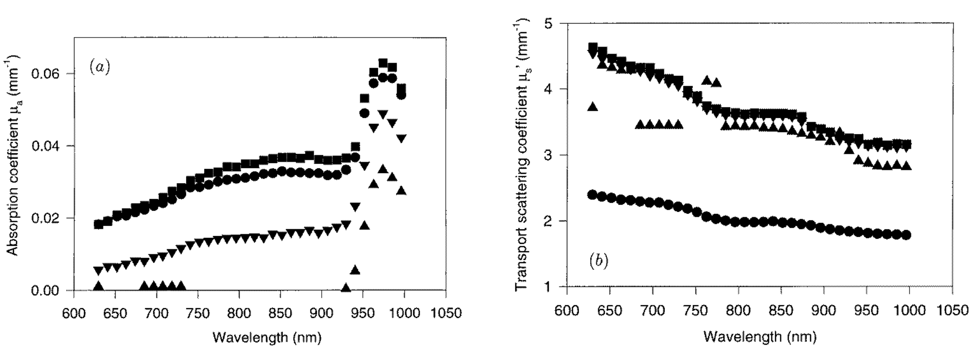

Figure 6. Optical coefficients of microsphere solutions: (*a*) absorption and (*b*) transport scattering. Calculated coefficients at 900 nm were sample 1 (●) $\mu_a = 0.043$ mm^{-1}, $\mu_s' = 1.0$ mm^{-1}; sample 2 (■) $\mu_a = 0.043$ mm^{-1}, $\mu_s' = 2.0$ mm^{-1}; sample 3 (▲, no dye) $\mu_a = 0$ mm^{-1}, $\mu_s' = 2.0$ mm^{-1}; sample 4 (▼) $\mu_a = 0.021$ mm^{-1}, $\mu_s' = 2.0$ mm^{-1}.

4. Evaluation against optical phantoms

The validity of the methodology was estimated by measuring two types of phantoms commonly used to mimic tissue: an aqueous solution of 1.27μm polystyrene microspheres

Table 1. Constituents of solid phantoms.

Scattering (arbitrary units)	Absorption (arbitrary units)		
	4	2	1
4	Phantom c	Phantom f	Phantom i
2	Phantom b	Phantom e	Phantom h
1	Phantom a	Phantom d	Phantom g

and infrared dye (S109564 Zeneca, Manchester, UK), and a solid epoxy resin phantom (Firbank and Delpy 1993). Figures 6(*a*) and (*b*) show the measured optical properties of four microsphere solutions along with their calculated equivalents. The absorption coefficient of the solutions was estimated from the known absorption spectra of pure dye and the transport scattering coefficient ($\mu_s' = \mu_s(1-g)$) calculated from Mie theory. It can be seen that the technique significantly overestimates the absolute transport scattering coefficient and underestimates absolute absorption coefficient. Ignoring the spectra for zero dye concentration (i.e. pure water absorption), the relative values of both μ_s' and μ_a are significantly better than the absolute values. The two plots in figure 6 marked ▼ and ■ represent identical μ_s' with a doubling in dye concentration. Here the addition of absorber led to minimal crosstalk in μ_s'. The observed μ_s' changed by less than 2% while μ_a doubled at wavelengths where water absorption was insignificant. Similarly the two plots in figure 6 marked ● and ■ represent the same μ_a at double the μ_s'. Here the crosstalk is only minimal below 750 nm and degrades to a 15% change in μ_a where no change was expected at 900 nm. This evaluation leads to the conclusion that the absolute values are not trustworthy but that the relative values in μ_s' are accurate to within 2% for $0.005 < \mu_a < 0.06$ mm^{-1} while the relative values in μ_a are accurate to within 15% for $1 < \mu_s' < 2.5$ mm^{-1}. The reasons for the discrepancy in absolute values are unclear. However, the solutions were mounted in the sample holders designed for tissue between glass coverslips and it may be that there are significant edge effects causing the polystyrene spheres to be non-uniformally distributed in depth due to electrostatic forces. The results would be consistent with the microspheres being concentrated in a thinner layer presenting a higher μ_s' and resulting in higher reflectivity with similar transmission. If this is the case, then it comes as no surprise that the near zero absorption data returned spurious values.

The uncertainty in the absolute value evaluation led to a second evaluation based upon solid phantoms where edge effects could be eliminated. A set of nine solid phantoms was measured using the same inversion technique. The phantoms (Firbank and Delpy 1993) were designed to nominally contain 1:2:4 ratios of scattering and absorbing components as shown in table 1, and were composed of 'Araldite' epoxy resin (Ciba Polymers, Cambridge, UK) containing titanium dioxide (0.3μm) particles and Projet 900NP dye (Zeneca, Manchester, UK). At each combination of properties, sufficient resin was mixed to make both multiple 16 mm diameter by 2 mm thick discs plus a 200 × 100 × 18 mm thick slab.

Discs of the phantoms were placed in the tissue sample holders, and reflectance and transmittance measured using the method described for tissue and the same inversion procedure was followed. The temporal response of the same material was measured on a large slab of phantom using picosecond pulses from a Ti:sapphire laser tuned to 799 nm and a streak camera following the method of Hebden (1995). The resulting TPSF was fitted using a least-squares algorithm to the diffusion equation applied to a slab (Patterson *et al* 1989) to produce the transport scattering and absorption coefficients. The extrapolated boundary condition was used.

Figure 7 shows the absorption and transport scattering coefficients obtained from the Monte Carlo inversion technique for the discs and the 'gold standard' analysis of the properties determined from the temporal point spread function (TPSF) of the larger slab. Two discs were measured from each phantom, and their clustering gives an indication not only of the reproducibility of the measurements but also the homogeneity of the phantom (phantom f in particular was poorly mixed, and variations in the phantom could be seen by eye when illuminated with a light box).

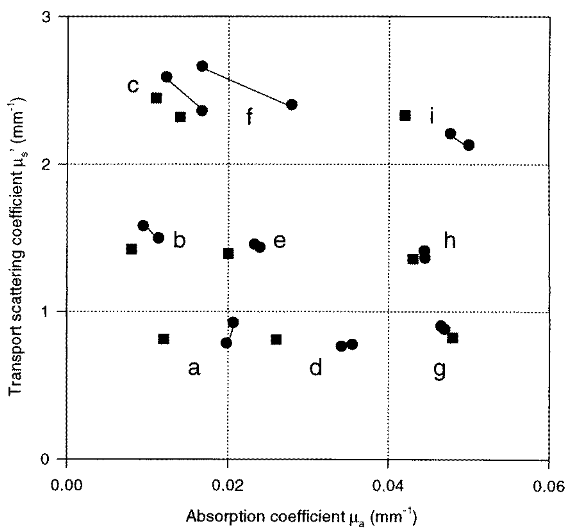

Figure 7. Absorption and transport scattering coefficients at 800 nm of the nine solid phantoms, calculated from (●) Monte Carlo inversion technique (two independent measurements per phantom) and (■) diffusion equation fit to TPSF (one measurement per phantom).

The correspondence between the transport scattering values predicted by the two techniques is excellent, being less than 5%, but there are systematic differences in the absorption coefficient especially at low values, leading to errors up to 20%. This is the limit of agreement of the two methods under these experimental circumstances. The flatness of the contour lines seen in figure 4 suggests that the integrating sphere method amplifies any error in the reflectance, transmittance data significantly more in μ_a than μ_s'. Also this effect is worse at the lowest μ_a, μ_s' values and this phenomenon is the most likely explanation of this calibration result.

5. Optical properties of tissue

5.1. Dermis

Samples of dermis were obtained from four caucasian and one negroid subject. Figures 8(*a*) and (*b*) show the measured optical properties of the caucasian dermis. The absorption spectrum shows clear contributions from the tissue constituents deoxyhaemoglobin (765 nm) and water (965 nm). The scattering spectrum (shown as transport scattering coefficient for consistency) shows a gradual reduction in scattering with wavelength. The error bars represent the standard deviation of twelve tissue samples (three from each subject) measured twice, illuminated front and back in turn. This explains the reduction in the size of the error

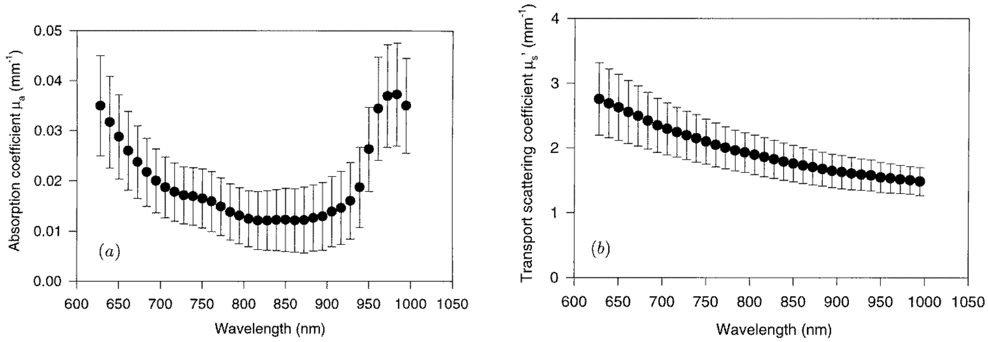

Figure 8. Optical coefficients of human caucasian dermis: (*a*) absorption coefficient and (*b*) transport scattering coefficient.

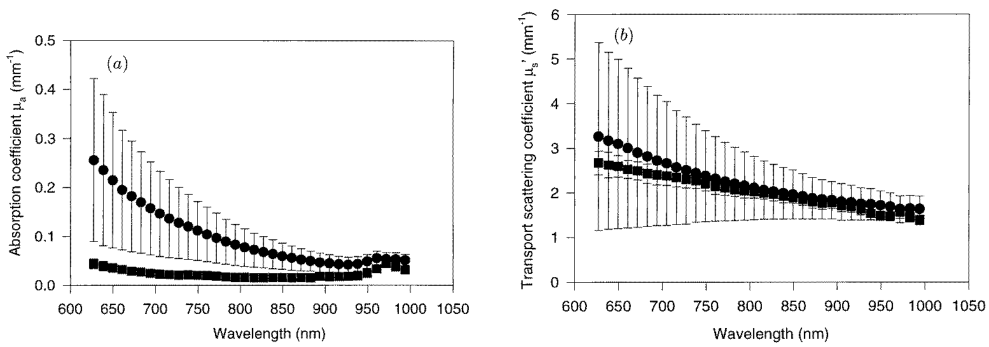

Figure 9. Optical coefficients of human dermis, caucasian (■) and negroid (●): (*a*) absorption coefficient and (*b*) transport scattering coefficient.

bars in the scattering coefficient with increasing wavelength. At shorter wavelengths, the differences in reflectivity between front and back illumination due to the heterogeneity (depth of absorbing layers) of the skin is more marked (Simpson *et al* 1997). Figure 4(*a*) shows that the scattering coefficient is more sensitive to changes in reflectance than the absorption coefficient in this range.

Figures 9(*a*) and (*b*) show the optical properties of the caucasian dermis plotted for comparison alongside data from a single negroid subject. The absorption coefficient for the negroid skin is an order of magnitude greater than the caucasian at visible wavelengths, reducing steadily with increasing wavelength. The size of the error bars again reflects the difference between front and back illumination of the sample and the smaller number of samples measured, three from a single subject. The scattering spectra show no significant differences between the negroid and caucasian skin.

5.2. Subdermis

Figures 10(*a*) and (*b*) show the results of the Monte Carlo inversion technique applied to measurements on subdermis from four caucasian subjects. No subdermis could be recovered from the negroid subject sample. The absorption spectrum again shows a feature at 765 nm but the shape is more suggestive of lipid than deoxyhaemoglobin. The major feature at 925 nm is due to lipid rather than water. The absorption and transport scattering coefficients are approximately half those of dermal tissue, with a slower rate of decrease in transport scattering coefficient at longer wavelengths. The error bars show none of the wavelength dependence seen in the dermal spectra because the subdermal tissue shows similar reflectance from both sides.

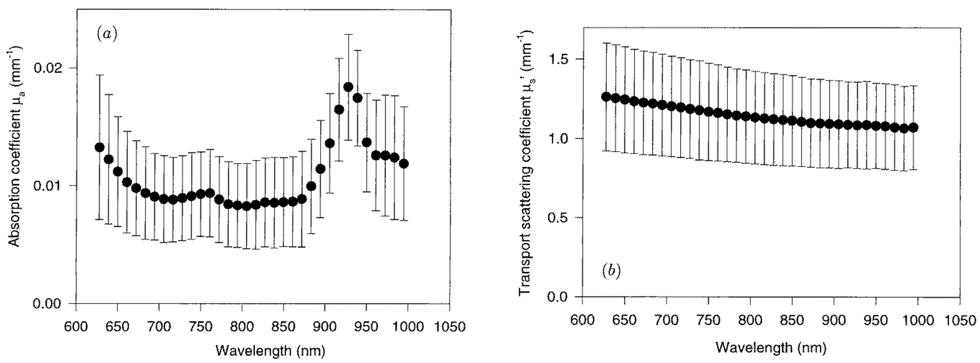

Figure 10. Optical coefficients of human caucasian subdermis: (*a*) absorption coefficient and (*b*) transport scattering coefficient.

5.3. Muscle

A single sample of abdominal muscle was recovered from one of the subjects and mounted so that the muscle fibres were perpendicular to the direction of illumination. The optical coefficients are presented in figures 11(*a*) and (*b*), and the spectra show the features seen in the previous tissue measurements. The absorption spectrum shows peaks due to deoxyhaemoglobin and water, and the scattering spectrum shows a gradual reduction in scattering with increasing wavelength.

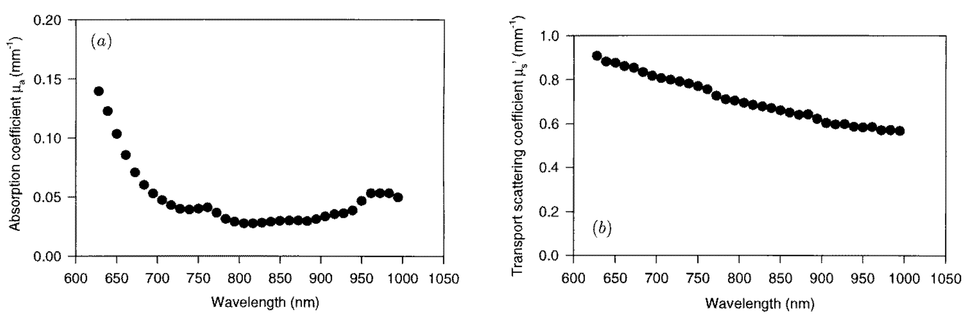

Figure 11. Optical coefficients of human abdominal muscle (*a*) absorption coefficient and (*b*) transport scattering coefficient.

6. Discussion

The optical coefficients obtained using the Monte Carlo inversion technique reflect the 'average' properties of the whole sample. For the subdermis, made up primarily of globular fat cells, averaging is not an important consideration. However, for 'dermis', which in reality are the stratum corneum, epidermis, dermis and vascular structures, average values are obviously a simplification. This is not necessarily a disadvantage when predicting the effects of skin and subcutaneous tissues on measurements of deeper tissue. The average coefficients are perhaps more useful than more exact measurements on thinner tissue layers in this situation. Additionally, Treweek and Barbenel (1996) found that at 633 nm the optical properties of dermis and epidermis were so similar that they could be considered as a single layer for all practical purposes.

The optical properties of skin have been reviewed elsewhere (van Gemert *et al* 1989, Graaff *et al* 1993) but there are contradictory values published for the absorption coefficient in particular. Graaff reanalysed the reflectance and transmittance data from Hardy *et al* (1956) and Jacques *et al* (1987) by comparison with a Monte Carlo model to produce the optical coefficients. He compared these with his own measurements of *in vivo* coefficients obtained from spatially resolved reflectance data fitted to a Monte Carlo model. The *in vivo* values for the optical coefficients were found to be an order of magnitude lower than the *ex vivo* measurements (table 2). Graaff suggested several explanations for this, the most plausible being that the reflectance and transmittance data of Hardy and Jacques, which were measured using a goniometer, were not accurate enough to predict the absorption coefficient successfully. Integrating sphere measurements should therefore give a better result. The recent values published by Treweek and Barbenel (1996) were also obtained using the integrated intensity from angular measurements and hence produce a poorly defined absorption coefficient.

Prahl (1988) measured the reflectance and transmittance of human dermis at 633 nm in an integrating sphere and calculated the optical coefficients using the adding-doubling method. However, separating the epidermis from the dermis required heating the sample to 55 °C in saline, which may have caused permanent changes in the tissue. Similar concerns exist when sample preparation involves freezing the tissue. Recently, Troy *et al* (1996) presented a single measurement of (previously frozen) dermis using a similar method. Again, however, there are inaccuracies in the absorption values due to corrections made for the double integrating sphere technique employed.

Table 2. Published values for the absorption and transport scattering coefficients of caucasian dermis.

Author	Wavelength (nm)	μ_a (mm^{-1})	μ_s' (mm^{-1})
Troy *et al* (1996)	749	0.0243 ± 0.019	2.314 ± 0.0753
Troy *et al* (1996)	789	0.0753 ± 0.006	2.276 ± 0.1294
Troy *et al* (1996)	836	0.0976 ± 0.015	1.594 ± 0.2158
Treweek and Barbenel (1996)	633	< 1.0	1.164 ($g = 0.97$)
Graaff *et al* (1993) (*in vivo*)	660	0.007–0.02	0.9–1.45
Graaff *et al* (1993) (analysis of data from Hardy *et al*)	700	0.27 ± 0.10	2.13 ± 0.37
Graaff *et al* (1993) (analysis of data from Jacques)	633	0.19 ± 0.06	2.38 ± 0.33
Prahl (1988)	633	0.15	5.02

The results presented here are consistent with previously reported values of transport scattering coefficient and show a monotonic decrease with increasing wavelength. Absorption values are lower than many reported values for the reasons explained above, but are consistent with the *in vivo* results of Graaff *et al* (1993). To estimate the system repeatability, a solid phantom sample was also measured on five occasions. The results suggest that the spread of results obtained for tissue transport scattering coefficient reflect

Table 3. Summary of absorption and transport scattering coefficients of various tissue types and phantoms from this paper.

Tissue	Wavelength (nm)	μ_a (mm^{-1})	μ_s' (mm^{-1})
Caucasian dermis ($n = 12$)	633	0.033 ± 0.009	2.73 ± 0.54
	700	0.019 ± 0.006	2.32 ± 0.41
	900	0.013 ± 0.007	1.63 ± 0.25
Negroid dermis ($n = 5$)	633	0.241 ± 0.153	3.21 ± 2.04
	700	0.149 ± 0.088	2.68 ± 1.41
	900	0.045 ± 0.018	1.81 ± 0.040
Subdermis ($n = 12$)	633	0.013 ± 0.005	1.26 ± 0.34
	700	0.009 ± 0.003	1.21 ± 0.32
	900	0.012 ± 0.004	1.08 ± 0.27
Muscle ($n = 1$)	633	0.121	0.89
	700	0.046	0.83
	900	0.032	0.59
Phantom ($n = 7$)	633	0.027 ± 0.007	2.14 ± 0.1

the inter- and intrasubject sample variability, whilst the variation in absorption coefficient between samples is within the system errors.

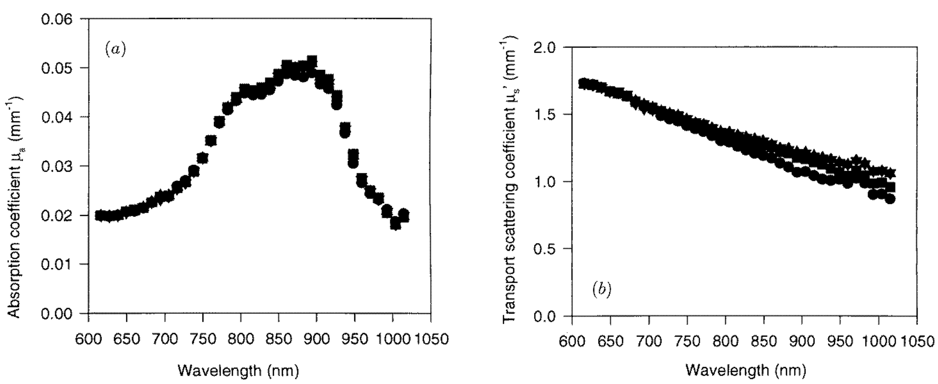

Figure 12. (a) Absorption and (b) transport scattering coefficient to show the effect of the value of g on the Monte Carlo inversion of a sample data set (phantom h); (●) $g = 0.80$, (■) $g = 0.85$, (▼) $g = 0.90$, (▲) $g = 0.95$.

Other possible sources of error are sample temperature, tissue storage and the scattering phase function used in the Monte Carlo model. The effects of temperature are discussed in a companion paper (Laufer *et al* 1988) and refrigerated storage of the sample for up to five days was found to have negligible effects on the measured optical properties (Simpson *et al* 1997). The sensitivity of the technique to the g of the scattering phase function was tested using the measurements on one of the solid calibration phantoms. Figure 12 shows the calculated value of (a) absorption coefficient and (b) transport scattering coefficient when the same reflectance and transmittance data were compared with four Monte Carlo data sets computed using g values from 0.8 to 0.95. A small ($< 1\%$) variation is observed in calculated transport scattering coefficient and negligible change in absorption coefficient. For this sample geometry and typical skin tissue values therefore, the technique is insensitive to the value of g used. This observation is also consistent with the results of Graaff *et al* (1993).

The overall accuracy of the absolute values obtained can be estimated by referring to the plot of solid phantom coefficients produced by the Monte Carlo inversion technique and analysis of the TPSF. The values of transport scattering coefficient are in extremely good agreement, within 5%. The absorption coefficients show greater variability, up to 20% errors, but the 'gold standard' TPSF method is itself more prone to errors in determining absorption coefficient compared with scattering coefficient.

7. Conclusions

The absorption and transport scattering coefficients of human dermis, subdermis and muscle were measured using a Monte Carlo inversion technique over a continuous wavelength range between 620 and 1000 nm. Absorption spectra showed the expected features due to tissue chromophores, and the transport scattering coefficient exhibited a steady decrease with increasing wavelength. The optical coefficients of subdermal tissue were found to be approximately half those of caucasian dermis. The single integrating sphere 'comparison' technique is shown to be a reliable method of measuring the light reflected and transmitted by tissue samples with sufficient accuracy to recover the absorption and transport scattering coefficient. Calibration against TPSF analysis has shown the degree of accuracy of the Monte Carlo model and the inversion technique. The accuracy is adequate for the purposes of providing optical coefficients of different tissue types in large complex structures such as the head, limbs and abdomen.

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